CLIENT	**:**	**NEOTRAFO SOLUTIONS INDIA PRIVATE LIMITED**
SITE	**:**	**LUCKNOW**
LOCATION	**:**	**LUCKNOW, INDIA**
NO.	**:**	**-**
DOCUMENT NO.	**:**	**DBR-R00**

R00	30-07-2026	Design report	R.S.R	H.A.B	H.A.B
Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

SALT
Design Services

TABLE OF CONTENTS

Sr. No	Contents	Page No.
1	Design Philosophy	3
2	Design Codes and Standards	3
3	Material Specifications Construction	4
4	Load Calculations	4
4.1	Dead Load	4
4.2	Live Load	5
5	Deflection Check	5
6	Primary Loads and Load Combinations	5
	STAAD output	9

SALT Design Services

1. Design Philosophy

This report presents the structural design of the fix MS Roof frame for Neotrafo at Lucknow, India. The supporting structure has been designed to safely resist the applicable dead loads (DL) and live loads (LL). The overall dimensions of the roof frame are **2.605 m × 4.145 m**. The frame is fabricated from mild steel. Wind load has not been considered in the design, as the structure is fully enclosed by the existing parapet walls

The complete frame is modelled as 3D frame and analysed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output.

2. DESIGN CODES AND STANDARDS

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
IS 875-Part I	For dead load calculation	--

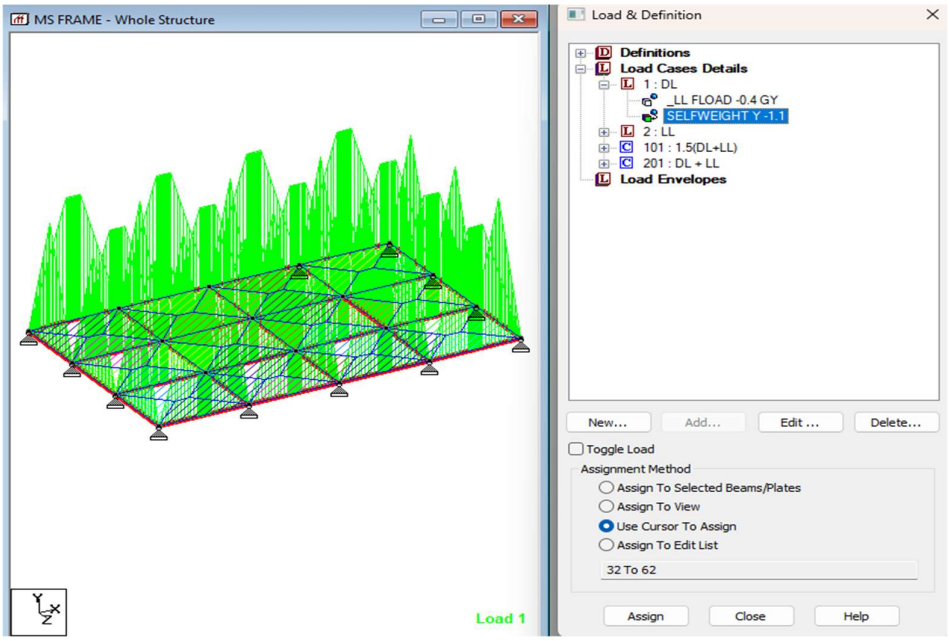
3. MATERIAL SPECIFICATION FOR CONSTRUCTION

Sr.no	Material	Specification	Minimum Strength
1	Mild steel	IS2062	Fy = 250 N/mm^2

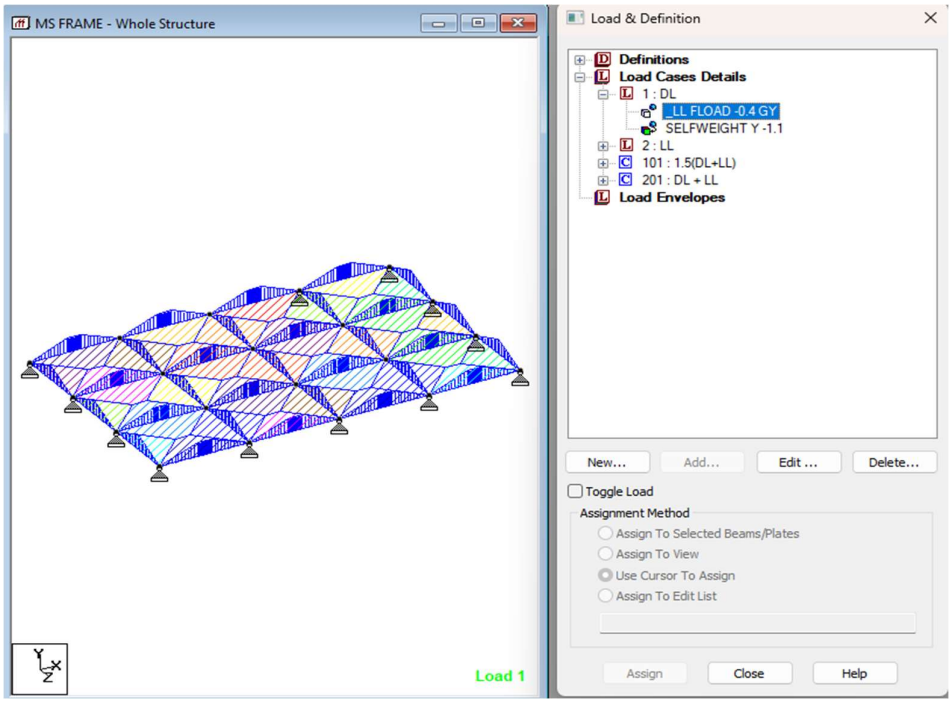
4. LOAD CALCULATIONS

4.1 DEAD LOAD

1.SELFWEIGHT FACTOR = 1.1

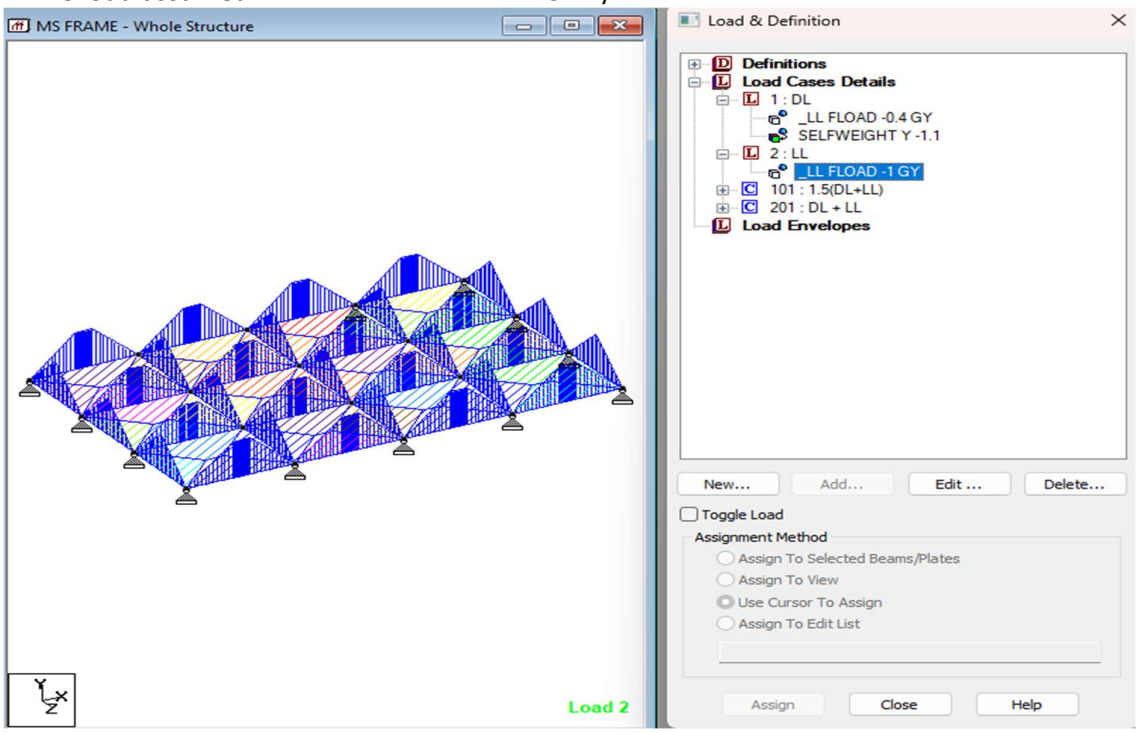


2. Dead load for 16mm thk. glass sheet assumed = 40 Kg/m^2 i.e. 0.40 kN/m^2



4.2 LIVE LOAD

1.Live load assumed = 1.5 kN/m^2



5. Deflection checks at building top in Y direction

Refer node no. –38 of Staad file for load combination 201

Max. Deflection in Y dir. = 16.557 mm < L/250 = (IS 800-2007 table 6)

.... Hence ok.

6. PRIMARY LOADS & LOAD COMBINATIONS

6.1 PRIMARY LOADS

Load 1 Dead Load (DL)

Load 2 Live Load (LL)

6.2 LOAD COMBINATIONS

Load Combination for strength: -

1.5 DL + 1.5 LL

Load Combination for serviceability: -

DL + LL

PAGE NO. 1

STAAD OUTPUT

```
**************************************************
*                                                *
*           STAAD.Pro 2023                       *
*           Version  23.00.03.25                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUL 29, 2026                *
*           Time=    18:21:36                    *
*                                                *

**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: E:\2041\MS frame\MS FRAME\MS FRAME.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 16-DEC-22
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 23 0 84.0002 5.00001; 24 0 84.0002 7.60502; 25 4.14501 84.0002 5.00001
     9. 26 4.14501 84.0002 7.60502; 27 1.03625 84.0002 5.00001
    10. 28 2.0725 84.0002 5.00001; 29 3.10876 84.0002 5.00001
    11. 30 1.03625 84.0002 7.60502; 31 2.0725 84.0002 7.60502
    12. 32 3.10876 84.0002 7.60502; 33 0 84.0002 5.86835; 34 0 84.0002 6.73668
    13. 35 4.14501 84.0002 5.86835; 36 4.14501 84.0002 6.73668
    14. 37 3.10876 84.0002 5.86835; 38 2.0725 84.0002 5.86835
    15. 39 1.03625 84.0002 5.86835; 40 3.10876 84.0002 6.73668
    16. 41 2.0725 84.0002 6.73668; 42 1.03625 84.0002 6.73668
    17. MEMBER INCIDENCES
    18. 32 23 33; 33 23 27; 34 24 30; 35 25 35; 36 27 28; 37 28 29; 38 29 25; 39 30 31
    19. 40 31 32; 41 32 26; 42 33 34; 43 34 24; 44 35 36; 45 36 26; 46 33 39; 47 34 42
    20. 48 27 39; 49 28 38; 50 29 37; 51 37 35; 52 38 37; 53 39 38; 54 40 36; 55 41 40
    21. 56 42 41; 57 42 30; 58 39 42; 59 41 31; 60 38 41; 61 40 32; 62 37 40
    22. START GROUP DEFINITION
    23. FLOOR
    24. _LL 32 TO 62
    25. END GROUP DEFINITION
    26. SUPPORTS
    27. 23 TO 26 29 TO 36 PINNED
    28. DEFINE MATERIAL START
    29. ISOTROPIC ALUMINUM
    30. E 6.89473E+07
    31. POISSON 0.33
    32. DENSITY 26.6017
    33. ALPHA 1.28E-05
    34. DAMP 0.03
    35. ISOTROPIC STEEL
    36. E 1.99947E+08
    37. POISSON 0.3
    38. DENSITY 76.8191
```

STAAD SPACE -- PAGE NO. 2

```
   39. ALPHA 6.5E-06
   40. DAMP 0.03
   41. TYPE STEEL
   42. STRENGTH RY 1.5 RT 1.2
   43. END DEFINE MATERIAL
   44. CONSTANTS
   45. MATERIAL STEEL ALL
   46. MEMBER PROPERTY APLAPOLLOTUBES
   47. 32 TO 45 48 TO 50 57 TO 62 TABLE ST 50X50X5.0SHS
   48. 46 47 51 TO 56 TABLE ST 50X50X5.0SHS
   49. DEFINE WIND LOAD

  *** NOTE: If any floor diaphragm is present in the model Wind Load definition
            should be defined after Floor Diaphragm definition. Otherwise wind
            load generation may be unsuccessful during analysis.

   50. TYPE 1 WIND 1
   51. INT 1.11 1.23 1.27 1.4 1.52 1.71 HEIG  10  20  25  30  50  100
   52. LOAD 1 LOADTYPE DEAD  TITLE DL
   53. FLOOR LOAD
   54. _LL FLOAD -0.4 GY
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

**NOTE** about FLOOR GROUP command
  All floor panels in a floor group should lie in a single plane.
  If not lying in a single plane break single FLOOR GROUP
  command into multiple FLOOR GROUP commands where each
  floor group lie in a single plane / height.

   55. SELFWEIGHT Y -1.1
   56. LOAD 2 LOADTYPE DEAD  TITLE LL
   57. FLOOR LOAD
   58. _LL FLOAD -1 GY
   59. LOAD COMB 101 1.5(DL+LL)
   60. 1 1.5 2 1.5
   61. LOAD COMB 201 DL + LL
   62. 1 1.0 2 1.0
   63. PERFORM ANALYSIS PRINT ALL
```

STAAD SPACE -- PAGE NO. 3

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	20	NUMBER OF MEMBERS	31
NUMBER OF PLATES	0	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	12

Using 64-bit analysis engine.

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 84
TOTAL LOAD COMBINATION CASES = 2 SO FAR.

STAAD SPACE -- PAGE NO. 4

LOADING 1 LOADTYPE DEAD TITLE DL

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
32				-0.0006 GY	0.04		
32				-0.0018 GY	0.08		
32				-0.0029 GY	0.14		
32				-0.0041 GY	0.19		
32				-0.0053 GY	0.25		
32				-0.0065 GY	0.30		
32				-0.0077 GY	0.35		
32				-0.0088 GY	0.41		
32				-0.0088 GY	0.46		
32				-0.0077 GY	0.51		
32				-0.0065 GY	0.57		
32				-0.0053 GY	0.62		
32				-0.0041 GY	0.68		
32				-0.0029 GY	0.73		
32				-0.0018 GY	0.78		
32				-0.0006 GY	0.83		
46				-0.0006 GY	0.04		
46				-0.0018 GY	0.08		
46				-0.0029 GY	0.14		
46				-0.0041 GY	0.19		
46				-0.0053 GY	0.25		
46				-0.0065 GY	0.30		
46				-0.0077 GY	0.35		
46				-0.0088 GY	0.41		
46	-0.1737 GY	0.43	0.60				
46				-0.0088 GY	0.63		
46				-0.0077 GY	0.68		
46				-0.0065 GY	0.74		
46				-0.0053 GY	0.79		
46				-0.0041 GY	0.85		
46				-0.0029 GY	0.90		
46				-0.0018 GY	0.95		
46				-0.0006 GY	1.00		
48				-0.0006 GY	0.04		
48				-0.0018 GY	0.08		
48				-0.0029 GY	0.14		
48				-0.0041 GY	0.19		
48				-0.0053 GY	0.25		
48				-0.0065 GY	0.30		
48				-0.0077 GY	0.35		
48				-0.0088 GY	0.41		
48				-0.0088 GY	0.46		
48				-0.0077 GY	0.51		
48				-0.0065 GY	0.57		
48				-0.0053 GY	0.62		
48				-0.0041 GY	0.68		

48			-0.0029	GY	0.73
48			-0.0018	GY	0.78
48			-0.0006	GY	0.83
33			-0.0006	GY	0.04
33			-0.0018	GY	0.08
33			-0.0029	GY	0.14
33			-0.0041	GY	0.19
33			-0.0053	GY	0.25
33			-0.0065	GY	0.30
33			-0.0077	GY	0.35
33			-0.0088	GY	0.41
33	-0.1737	GY 0.43	0.60		
33			-0.0088	GY	0.63
33			-0.0077	GY	0.68
33			-0.0065	GY	0.74
33			-0.0053	GY	0.79
33			-0.0041	GY	0.85
33			-0.0029	GY	0.90
33			-0.0018	GY	0.95
33			-0.0006	GY	1.00
34			-0.0006	GY	0.04
34			-0.0018	GY	0.08
34			-0.0029	GY	0.14
34			-0.0041	GY	0.19
34			-0.0053	GY	0.25
34			-0.0065	GY	0.30
34			-0.0077	GY	0.35
34			-0.0088	GY	0.41
34	-0.1737	GY 0.43	0.60		
34			-0.0088	GY	0.63
34			-0.0077	GY	0.68
34			-0.0065	GY	0.74
34			-0.0053	GY	0.79
34			-0.0041	GY	0.85
34			-0.0029	GY	0.90
34			-0.0018	GY	0.95
34			-0.0006	GY	1.00
57			-0.0006	GY	0.04
57			-0.0018	GY	0.08
57			-0.0029	GY	0.14
57			-0.0041	GY	0.19
57			-0.0053	GY	0.25
57			-0.0065	GY	0.30
57			-0.0077	GY	0.35
57			-0.0088	GY	0.41
57			-0.0088	GY	0.46
57			-0.0077	GY	0.51
57			-0.0065	GY	0.57
57			-0.0053	GY	0.62
57			-0.0041	GY	0.68
57			-0.0029	GY	0.73
57			-0.0018	GY	0.78
57			-0.0006	GY	0.83
47			-0.0006	GY	0.04
47			-0.0018	GY	0.08
47			-0.0029	GY	0.14

STAAD SPACE -- PAGE NO. 6

47			-0.0041 GY	0.19
47			-0.0053 GY	0.25
47			-0.0065 GY	0.30
47			-0.0077 GY	0.35
47			-0.0088 GY	0.41
47	-0.1737 GY	0.43	0.60	
47			-0.0088 GY	0.63
47			-0.0077 GY	0.68
47			-0.0065 GY	0.74
47			-0.0053 GY	0.79
47			-0.0041 GY	0.85
47			-0.0029 GY	0.90
47			-0.0018 GY	0.95
47			-0.0006 GY	1.00
43			-0.0006 GY	0.04
43			-0.0018 GY	0.08
43			-0.0029 GY	0.14
43			-0.0041 GY	0.19
43			-0.0053 GY	0.25
43			-0.0065 GY	0.30
43			-0.0077 GY	0.35
43			-0.0088 GY	0.41
43			-0.0088 GY	0.46
43			-0.0077 GY	0.51
43			-0.0065 GY	0.57
43			-0.0053 GY	0.62
43			-0.0041 GY	0.68
43			-0.0029 GY	0.73
43			-0.0018 GY	0.78
43			-0.0006 GY	0.83
35			-0.0006 GY	0.04
35			-0.0018 GY	0.08
35			-0.0029 GY	0.14
35			-0.0041 GY	0.19
35			-0.0053 GY	0.25
35			-0.0065 GY	0.30
35			-0.0077 GY	0.35
35			-0.0088 GY	0.41
35			-0.0088 GY	0.46
35			-0.0077 GY	0.51
35			-0.0065 GY	0.57
35			-0.0053 GY	0.62
35			-0.0041 GY	0.68
35			-0.0029 GY	0.73
35			-0.0018 GY	0.78
35			-0.0006 GY	0.83
51			-0.0006 GY	0.04
51			-0.0018 GY	0.08
51			-0.0029 GY	0.14
51			-0.0041 GY	0.19
51			-0.0053 GY	0.25
51			-0.0065 GY	0.30
51			-0.0077 GY	0.35
51			-0.0088 GY	0.41
51	-0.1737 GY	0.43	0.60	
51			-0.0088 GY	0.63

STAAD SPACE -- PAGE NO. 7

51			-0.0077 GY	0.68
51			-0.0065 GY	0.74
51			-0.0053 GY	0.79
51			-0.0041 GY	0.85
51			-0.0029 GY	0.90
51			-0.0018 GY	0.95
51			-0.0006 GY	1.00
50			-0.0006 GY	0.04
50			-0.0018 GY	0.08
50			-0.0029 GY	0.14
50			-0.0041 GY	0.19
50			-0.0053 GY	0.25
50			-0.0065 GY	0.30
50			-0.0077 GY	0.35
50			-0.0088 GY	0.41
50			-0.0088 GY	0.46
50			-0.0077 GY	0.51
50			-0.0065 GY	0.57
50			-0.0053 GY	0.62
50			-0.0041 GY	0.68
50			-0.0029 GY	0.73
50			-0.0018 GY	0.78
50			-0.0006 GY	0.83
38			-0.0006 GY	0.04
38			-0.0018 GY	0.08
38			-0.0029 GY	0.14
38			-0.0041 GY	0.19
38			-0.0053 GY	0.25
38			-0.0065 GY	0.30
38			-0.0077 GY	0.35
38			-0.0088 GY	0.41
38	-0.1737 GY	0.43	0.60	
38			-0.0088 GY	0.63
38			-0.0077 GY	0.68
38			-0.0065 GY	0.74
38			-0.0053 GY	0.79
38			-0.0041 GY	0.85
38			-0.0029 GY	0.90
38			-0.0018 GY	0.95
38			-0.0006 GY	1.00
36			-0.0006 GY	0.04
36			-0.0018 GY	0.08
36			-0.0029 GY	0.14
36			-0.0041 GY	0.19
36			-0.0053 GY	0.25
36			-0.0065 GY	0.30
36			-0.0077 GY	0.35
36			-0.0088 GY	0.41
36	-0.1737 GY	0.43	0.60	
36			-0.0088 GY	0.63
36			-0.0077 GY	0.68
36			-0.0065 GY	0.74
36			-0.0053 GY	0.79
36			-0.0041 GY	0.85
36			-0.0029 GY	0.90
36			-0.0018 GY	0.95

STAAD SPACE -- PAGE NO. 8

36			-0.0006 GY	1.00
49			-0.0006 GY	0.04
49			-0.0018 GY	0.08
49			-0.0029 GY	0.14
49			-0.0041 GY	0.19
49			-0.0053 GY	0.25
49			-0.0065 GY	0.30
49			-0.0077 GY	0.35
49			-0.0088 GY	0.41
49			-0.0088 GY	0.46
49			-0.0077 GY	0.51
49			-0.0065 GY	0.57
49			-0.0053 GY	0.62
49			-0.0041 GY	0.68
49			-0.0029 GY	0.73
49			-0.0018 GY	0.78
49			-0.0006 GY	0.83
53			-0.0006 GY	0.04
53			-0.0018 GY	0.08
53			-0.0029 GY	0.14
53			-0.0041 GY	0.19
53			-0.0053 GY	0.25
53			-0.0065 GY	0.30
53			-0.0077 GY	0.35
53			-0.0088 GY	0.41
53	-0.1737 GY	0.43	0.60	
53			-0.0088 GY	0.63
53			-0.0077 GY	0.68
53			-0.0065 GY	0.74
53			-0.0053 GY	0.79
53			-0.0041 GY	0.85
53			-0.0029 GY	0.90
53			-0.0018 GY	0.95
53			-0.0006 GY	1.00
48			-0.0006 GY	0.04
48			-0.0018 GY	0.08
48			-0.0029 GY	0.14
48			-0.0041 GY	0.19
48			-0.0053 GY	0.25
48			-0.0065 GY	0.30
48			-0.0077 GY	0.35
48			-0.0088 GY	0.41
48			-0.0088 GY	0.46
48			-0.0077 GY	0.51
48			-0.0065 GY	0.57
48			-0.0053 GY	0.62
48			-0.0041 GY	0.68
48			-0.0029 GY	0.73
48			-0.0018 GY	0.78
48			-0.0006 GY	0.83
37			-0.0006 GY	0.04
37			-0.0018 GY	0.08
37			-0.0029 GY	0.14
37			-0.0041 GY	0.19
37			-0.0053 GY	0.25
37			-0.0065 GY	0.30

37			-0.0077 GY	0.35	
37			-0.0088 GY	0.41	
37	-0.1737 GY	0.43	0.60		
37			-0.0088 GY	0.63	
37			-0.0077 GY	0.68	
37			-0.0065 GY	0.74	
37			-0.0053 GY	0.79	
37			-0.0041 GY	0.85	
37			-0.0029 GY	0.90	
37			-0.0018 GY	0.95	
37			-0.0006 GY	1.00	
50			-0.0006 GY	0.04	
50			-0.0018 GY	0.08	
50			-0.0029 GY	0.14	
50			-0.0041 GY	0.19	
50			-0.0053 GY	0.25	
50			-0.0065 GY	0.30	
50			-0.0077 GY	0.35	
50			-0.0088 GY	0.41	
50			-0.0088 GY	0.46	
50			-0.0077 GY	0.51	
50			-0.0065 GY	0.57	
50			-0.0053 GY	0.62	
50			-0.0041 GY	0.68	
50			-0.0029 GY	0.73	
50			-0.0018 GY	0.78	
50			-0.0006 GY	0.83	
52			-0.0006 GY	0.04	
52			-0.0018 GY	0.08	
52			-0.0029 GY	0.14	
52			-0.0041 GY	0.19	
52			-0.0053 GY	0.25	
52			-0.0065 GY	0.30	
52			-0.0077 GY	0.35	
52			-0.0088 GY	0.41	
52	-0.1737 GY	0.43	0.60		
52			-0.0088 GY	0.63	
52			-0.0077 GY	0.68	
52			-0.0065 GY	0.74	
52			-0.0053 GY	0.79	
52			-0.0041 GY	0.85	
52			-0.0029 GY	0.90	
52			-0.0018 GY	0.95	
52			-0.0006 GY	1.00	
49			-0.0006 GY	0.04	
49			-0.0018 GY	0.08	
49			-0.0029 GY	0.14	
49			-0.0041 GY	0.19	
49			-0.0053 GY	0.25	
49			-0.0065 GY	0.30	
49			-0.0077 GY	0.35	
49			-0.0088 GY	0.41	
49			-0.0088 GY	0.46	
49			-0.0077 GY	0.51	
49			-0.0065 GY	0.57	
49			-0.0053 GY	0.62	

STAAD SPACE -- PAGE NO. 10

49			-0.0041 GY	0.68	
49			-0.0029 GY	0.73	
49			-0.0018 GY	0.78	
49			-0.0006 GY	0.83	
39			-0.0006 GY	0.04	
39			-0.0018 GY	0.08	
39			-0.0029 GY	0.14	
39			-0.0041 GY	0.19	
39			-0.0053 GY	0.25	
39			-0.0065 GY	0.30	
39			-0.0077 GY	0.35	
39			-0.0088 GY	0.41	
39	-0.1737 GY	0.43	0.60		
39			-0.0088 GY	0.63	
39			-0.0077 GY	0.68	
39			-0.0065 GY	0.74	
39			-0.0053 GY	0.79	
39			-0.0041 GY	0.85	
39			-0.0029 GY	0.90	
39			-0.0018 GY	0.95	
39			-0.0006 GY	1.00	
59			-0.0006 GY	0.04	
59			-0.0018 GY	0.08	
59			-0.0029 GY	0.14	
59			-0.0041 GY	0.19	
59			-0.0053 GY	0.25	
59			-0.0065 GY	0.30	
59			-0.0077 GY	0.35	
59			-0.0088 GY	0.41	
59			-0.0088 GY	0.46	
59			-0.0077 GY	0.51	
59			-0.0065 GY	0.57	
59			-0.0053 GY	0.62	
59			-0.0041 GY	0.68	
59			-0.0029 GY	0.73	
59			-0.0018 GY	0.78	
59			-0.0006 GY	0.83	
56			-0.0006 GY	0.04	
56			-0.0018 GY	0.08	
56			-0.0029 GY	0.14	
56			-0.0041 GY	0.19	
56			-0.0053 GY	0.25	
56			-0.0065 GY	0.30	
56			-0.0077 GY	0.35	
56			-0.0088 GY	0.41	
56	-0.1737 GY	0.43	0.60		
56			-0.0088 GY	0.63	
56			-0.0077 GY	0.68	
56			-0.0065 GY	0.74	
56			-0.0053 GY	0.79	
56			-0.0041 GY	0.85	
56			-0.0029 GY	0.90	
56			-0.0018 GY	0.95	
56			-0.0006 GY	1.00	
57			-0.0006 GY	0.04	
57			-0.0018 GY	0.08	

57			-0.0029 GY	0.14	
57			-0.0041 GY	0.19	
57			-0.0053 GY	0.25	
57			-0.0065 GY	0.30	
57			-0.0077 GY	0.35	
57			-0.0088 GY	0.41	
57			-0.0088 GY	0.46	
57			-0.0077 GY	0.51	
57			-0.0065 GY	0.57	
57			-0.0053 GY	0.62	
57			-0.0041 GY	0.68	
57			-0.0029 GY	0.73	
57			-0.0018 GY	0.78	
57			-0.0006 GY	0.83	
40			-0.0006 GY	0.04	
40			-0.0018 GY	0.08	
40			-0.0029 GY	0.14	
40			-0.0041 GY	0.19	
40			-0.0053 GY	0.25	
40			-0.0065 GY	0.30	
40			-0.0077 GY	0.35	
40			-0.0088 GY	0.41	
40	-0.1737 GY	0.43	0.60		
40			-0.0088 GY	0.63	
40			-0.0077 GY	0.68	
40			-0.0065 GY	0.74	
40			-0.0053 GY	0.79	
40			-0.0041 GY	0.85	
40			-0.0029 GY	0.90	
40			-0.0018 GY	0.95	
40			-0.0006 GY	1.00	
61			-0.0006 GY	0.04	
61			-0.0018 GY	0.08	
61			-0.0029 GY	0.14	
61			-0.0041 GY	0.19	
61			-0.0053 GY	0.25	
61			-0.0065 GY	0.30	
61			-0.0077 GY	0.35	
61			-0.0088 GY	0.41	
61			-0.0088 GY	0.46	
61			-0.0077 GY	0.51	
61			-0.0065 GY	0.57	
61			-0.0053 GY	0.62	
61			-0.0041 GY	0.68	
61			-0.0029 GY	0.73	
61			-0.0018 GY	0.78	
61			-0.0006 GY	0.83	
55			-0.0006 GY	0.04	
55			-0.0018 GY	0.08	
55			-0.0029 GY	0.14	
55			-0.0041 GY	0.19	
55			-0.0053 GY	0.25	
55			-0.0065 GY	0.30	
55			-0.0077 GY	0.35	
55			-0.0088 GY	0.41	
55	-0.1737 GY	0.43	0.60		

55			-0.0088 GY	0.63	
55			-0.0077 GY	0.68	
55			-0.0065 GY	0.74	
55			-0.0053 GY	0.79	
55			-0.0041 GY	0.85	
55			-0.0029 GY	0.90	
55			-0.0018 GY	0.95	
55			-0.0006 GY	1.00	
59			-0.0006 GY	0.04	
59			-0.0018 GY	0.08	
59			-0.0029 GY	0.14	
59			-0.0041 GY	0.19	
59			-0.0053 GY	0.25	
59			-0.0065 GY	0.30	
59			-0.0077 GY	0.35	
59			-0.0088 GY	0.41	
59			-0.0088 GY	0.46	
59			-0.0077 GY	0.51	
59			-0.0065 GY	0.57	
59			-0.0053 GY	0.62	
59			-0.0041 GY	0.68	
59			-0.0029 GY	0.73	
59			-0.0018 GY	0.78	
59			-0.0006 GY	0.83	
41			-0.0006 GY	0.04	
41			-0.0018 GY	0.08	
41			-0.0029 GY	0.14	
41			-0.0041 GY	0.19	
41			-0.0053 GY	0.25	
41			-0.0065 GY	0.30	
41			-0.0077 GY	0.35	
41			-0.0088 GY	0.41	
41	-0.1737 GY	0.43	0.60		
41			-0.0088 GY	0.63	
41			-0.0077 GY	0.68	
41			-0.0065 GY	0.74	
41			-0.0053 GY	0.79	
41			-0.0041 GY	0.85	
41			-0.0029 GY	0.90	
41			-0.0018 GY	0.95	
41			-0.0006 GY	1.00	
45			-0.0006 GY	0.04	
45			-0.0018 GY	0.08	
45			-0.0029 GY	0.14	
45			-0.0041 GY	0.19	
45			-0.0053 GY	0.25	
45			-0.0065 GY	0.30	
45			-0.0077 GY	0.35	
45			-0.0088 GY	0.41	
45			-0.0088 GY	0.46	
45			-0.0077 GY	0.51	
45			-0.0065 GY	0.57	
45			-0.0053 GY	0.62	
45			-0.0041 GY	0.68	
45			-0.0029 GY	0.73	
45			-0.0018 GY	0.78	

```
 45                          -0.0006 GY   0.83
 54                          -0.0006 GY   0.04
 54                          -0.0018 GY   0.08
 54                          -0.0029 GY   0.14
 54                          -0.0041 GY   0.19
 54                          -0.0053 GY   0.25
 54                          -0.0065 GY   0.30
 54                          -0.0077 GY   0.35
 54                          -0.0088 GY   0.41
 54    -0.1737 GY   0.43    0.60
 54                          -0.0088 GY   0.63
 54                          -0.0077 GY   0.68
 54                          -0.0065 GY   0.74
 54                          -0.0053 GY   0.79
 54                          -0.0041 GY   0.85
 54                          -0.0029 GY   0.90
 54                          -0.0018 GY   0.95
 54                          -0.0006 GY   1.00
 61                          -0.0006 GY   0.04
 61                          -0.0018 GY   0.08
 61                          -0.0029 GY   0.14
 61                          -0.0041 GY   0.19
 61                          -0.0053 GY   0.25
 61                          -0.0065 GY   0.30
 61                          -0.0077 GY   0.35
 61                          -0.0088 GY   0.41
 61                          -0.0088 GY   0.46
 61                          -0.0077 GY   0.51
 61                          -0.0065 GY   0.57
 61                          -0.0053 GY   0.62
 61                          -0.0041 GY   0.68
 61                          -0.0029 GY   0.73
 61                          -0.0018 GY   0.78
 61                          -0.0006 GY   0.83
 42                          -0.0006 GY   0.04
 42                          -0.0018 GY   0.08
 42                          -0.0029 GY   0.14
 42                          -0.0041 GY   0.19
 42                          -0.0053 GY   0.25
 42                          -0.0065 GY   0.30
 42                          -0.0077 GY   0.35
 42                          -0.0088 GY   0.41
 42                          -0.0088 GY   0.46
 42                          -0.0077 GY   0.51
 42                          -0.0065 GY   0.57
 42                          -0.0053 GY   0.62
 42                          -0.0041 GY   0.68
 42                          -0.0029 GY   0.73
 42                          -0.0018 GY   0.78
 42                          -0.0006 GY   0.83
 47                          -0.0006 GY   0.04
 47                          -0.0018 GY   0.08
 47                          -0.0029 GY   0.14
 47                          -0.0041 GY   0.19
 47                          -0.0053 GY   0.25
 47                          -0.0065 GY   0.30
```

47			-0.0077	GY	0.35
47			-0.0088	GY	0.41
47	-0.1737	GY	0.43	0.60	
47			-0.0088	GY	0.63
47			-0.0077	GY	0.68
47			-0.0065	GY	0.74
47			-0.0053	GY	0.79
47			-0.0041	GY	0.85
47			-0.0029	GY	0.90
47			-0.0018	GY	0.95
47			-0.0006	GY	1.00
58			-0.0006	GY	0.04
58			-0.0018	GY	0.08
58			-0.0029	GY	0.14
58			-0.0041	GY	0.19
58			-0.0053	GY	0.25
58			-0.0065	GY	0.30
58			-0.0077	GY	0.35
58			-0.0088	GY	0.41
58			-0.0088	GY	0.46
58			-0.0077	GY	0.51
58			-0.0065	GY	0.57
58			-0.0053	GY	0.62
58			-0.0041	GY	0.68
58			-0.0029	GY	0.73
58			-0.0018	GY	0.78
58			-0.0006	GY	0.83
46			-0.0006	GY	0.04
46			-0.0018	GY	0.08
46			-0.0029	GY	0.14
46			-0.0041	GY	0.19
46			-0.0053	GY	0.25
46			-0.0065	GY	0.30
46			-0.0077	GY	0.35
46			-0.0088	GY	0.41
46	-0.1737	GY	0.43	0.60	
46			-0.0088	GY	0.63
46			-0.0077	GY	0.68
46			-0.0065	GY	0.74
46			-0.0053	GY	0.79
46			-0.0041	GY	0.85
46			-0.0029	GY	0.90
46			-0.0018	GY	0.95
46			-0.0006	GY	1.00
44			-0.0006	GY	0.04
44			-0.0018	GY	0.08
44			-0.0029	GY	0.14
44			-0.0041	GY	0.19
44			-0.0053	GY	0.25
44			-0.0065	GY	0.30
44			-0.0077	GY	0.35
44			-0.0088	GY	0.41
44			-0.0088	GY	0.46
44			-0.0077	GY	0.51
44			-0.0065	GY	0.57
44			-0.0053	GY	0.62

STAAD SPACE -- PAGE NO. 15

44			-0.0041 GY	0.68	
44			-0.0029 GY	0.73	
44			-0.0018 GY	0.78	
44			-0.0006 GY	0.83	
54			-0.0006 GY	0.04	
54			-0.0018 GY	0.08	
54			-0.0029 GY	0.14	
54			-0.0041 GY	0.19	
54			-0.0053 GY	0.25	
54			-0.0065 GY	0.30	
54			-0.0077 GY	0.35	
54			-0.0088 GY	0.41	
54	-0.1737 GY	0.43	0.60		
54			-0.0088 GY	0.63	
54			-0.0077 GY	0.68	
54			-0.0065 GY	0.74	
54			-0.0053 GY	0.79	
54			-0.0041 GY	0.85	
54			-0.0029 GY	0.90	
54			-0.0018 GY	0.95	
54			-0.0006 GY	1.00	
62			-0.0006 GY	0.04	
62			-0.0018 GY	0.08	
62			-0.0029 GY	0.14	
62			-0.0041 GY	0.19	
62			-0.0053 GY	0.25	
62			-0.0065 GY	0.30	
62			-0.0077 GY	0.35	
62			-0.0088 GY	0.41	
62			-0.0088 GY	0.46	
62			-0.0077 GY	0.51	
62			-0.0065 GY	0.57	
62			-0.0053 GY	0.62	
62			-0.0041 GY	0.68	
62			-0.0029 GY	0.73	
62			-0.0018 GY	0.78	
62			-0.0006 GY	0.83	
51			-0.0006 GY	0.04	
51			-0.0018 GY	0.08	
51			-0.0029 GY	0.14	
51			-0.0041 GY	0.19	
51			-0.0053 GY	0.25	
51			-0.0065 GY	0.30	
51			-0.0077 GY	0.35	
51			-0.0088 GY	0.41	
51	-0.1737 GY	0.43	0.60		
51			-0.0088 GY	0.63	
51			-0.0077 GY	0.68	
51			-0.0065 GY	0.74	
51			-0.0053 GY	0.79	
51			-0.0041 GY	0.85	
51			-0.0029 GY	0.90	
51			-0.0018 GY	0.95	
51			-0.0006 GY	1.00	
52			-0.0006 GY	0.04	
52			-0.0018 GY	0.08	

52			-0.0029 GY	0.14
52			-0.0041 GY	0.19
52			-0.0053 GY	0.25
52			-0.0065 GY	0.30
52			-0.0077 GY	0.35
52			-0.0088 GY	0.41
52	-0.1737 GY	0.43	0.60	
52			-0.0088 GY	0.63
52			-0.0077 GY	0.68
52			-0.0065 GY	0.74
52			-0.0053 GY	0.79
52			-0.0041 GY	0.85
52			-0.0029 GY	0.90
52			-0.0018 GY	0.95
52			-0.0006 GY	1.00
62			-0.0006 GY	0.04
62			-0.0018 GY	0.08
62			-0.0029 GY	0.14
62			-0.0041 GY	0.19
62			-0.0053 GY	0.25
62			-0.0065 GY	0.30
62			-0.0077 GY	0.35
62			-0.0088 GY	0.41
62			-0.0088 GY	0.46
62			-0.0077 GY	0.51
62			-0.0065 GY	0.57
62			-0.0053 GY	0.62
62			-0.0041 GY	0.68
62			-0.0029 GY	0.73
62			-0.0018 GY	0.78
62			-0.0006 GY	0.83
55			-0.0006 GY	0.04
55			-0.0018 GY	0.08
55			-0.0029 GY	0.14
55			-0.0041 GY	0.19
55			-0.0053 GY	0.25
55			-0.0065 GY	0.30
55			-0.0077 GY	0.35
55			-0.0088 GY	0.41
55	-0.1737 GY	0.43	0.60	
55			-0.0088 GY	0.63
55			-0.0077 GY	0.68
55			-0.0065 GY	0.74
55			-0.0053 GY	0.79
55			-0.0041 GY	0.85
55			-0.0029 GY	0.90
55			-0.0018 GY	0.95
55			-0.0006 GY	1.00
60			-0.0006 GY	0.04
60			-0.0018 GY	0.08
60			-0.0029 GY	0.14
60			-0.0041 GY	0.19
60			-0.0053 GY	0.25
60			-0.0065 GY	0.30
60			-0.0077 GY	0.35
60			-0.0088 GY	0.41

60			-0.0088 GY	0.46	
60			-0.0077 GY	0.51	
60			-0.0065 GY	0.57	
60			-0.0053 GY	0.62	
60			-0.0041 GY	0.68	
60			-0.0029 GY	0.73	
60			-0.0018 GY	0.78	
60			-0.0006 GY	0.83	
53			-0.0006 GY	0.04	
53			-0.0018 GY	0.08	
53			-0.0029 GY	0.14	
53			-0.0041 GY	0.19	
53			-0.0053 GY	0.25	
53			-0.0065 GY	0.30	
53			-0.0077 GY	0.35	
53			-0.0088 GY	0.41	
53	-0.1737 GY	0.43	0.60		
53			-0.0088 GY	0.63	
53			-0.0077 GY	0.68	
53			-0.0065 GY	0.74	
53			-0.0053 GY	0.79	
53			-0.0041 GY	0.85	
53			-0.0029 GY	0.90	
53			-0.0018 GY	0.95	
53			-0.0006 GY	1.00	
60			-0.0006 GY	0.04	
60			-0.0018 GY	0.08	
60			-0.0029 GY	0.14	
60			-0.0041 GY	0.19	
60			-0.0053 GY	0.25	
60			-0.0065 GY	0.30	
60			-0.0077 GY	0.35	
60			-0.0088 GY	0.41	
60			-0.0088 GY	0.46	
60			-0.0077 GY	0.51	
60			-0.0065 GY	0.57	
60			-0.0053 GY	0.62	
60			-0.0041 GY	0.68	
60			-0.0029 GY	0.73	
60			-0.0018 GY	0.78	
60			-0.0006 GY	0.83	
56			-0.0006 GY	0.04	
56			-0.0018 GY	0.08	
56			-0.0029 GY	0.14	
56			-0.0041 GY	0.19	
56			-0.0053 GY	0.25	
56			-0.0065 GY	0.30	
56			-0.0077 GY	0.35	
56			-0.0088 GY	0.41	
56	-0.1737 GY	0.43	0.60		
56			-0.0088 GY	0.63	
56			-0.0077 GY	0.68	
56			-0.0065 GY	0.74	
56			-0.0053 GY	0.79	
56			-0.0041 GY	0.85	
56			-0.0029 GY	0.90	

56		-0.0018 GY	0.95
56		-0.0006 GY	1.00
58		-0.0006 GY	0.04
58		-0.0018 GY	0.08
58		-0.0029 GY	0.14
58		-0.0041 GY	0.19
58		-0.0053 GY	0.25
58		-0.0065 GY	0.30
58		-0.0077 GY	0.35
58		-0.0088 GY	0.41
58		-0.0088 GY	0.46
58		-0.0077 GY	0.51
58		-0.0065 GY	0.57
58		-0.0053 GY	0.62
58		-0.0041 GY	0.68
58		-0.0029 GY	0.73
58		-0.0018 GY	0.78
58		-0.0006 GY	0.83

SELFWEIGHT Y -1.100

ACTUAL WEIGHT OF THE STRUCTURE = 1.901 KN

LOADING 2 LOADTYPE DEAD TITLE LL

MEMBER LOAD - UNIT KN METE

MEMBER	UDL	L1	L2	CON	L	LIN1	LIN2
32				-0.0015 GY	0.04		
32				-0.0044 GY	0.08		
32				-0.0074 GY	0.14		
32				-0.0103 GY	0.19		
32				-0.0133 GY	0.25		
32				-0.0162 GY	0.30		
32				-0.0191 GY	0.35		
32				-0.0221 GY	0.41		
32				-0.0221 GY	0.46		
32				-0.0191 GY	0.51		
32				-0.0162 GY	0.57		
32				-0.0133 GY	0.62		
32				-0.0103 GY	0.68		
32				-0.0074 GY	0.73		
32				-0.0044 GY	0.78		
32				-0.0015 GY	0.83		
46				-0.0015 GY	0.04		
46				-0.0044 GY	0.08		
46				-0.0074 GY	0.14		
46				-0.0103 GY	0.19		
46				-0.0133 GY	0.25		
46				-0.0162 GY	0.30		
46				-0.0191 GY	0.35		
46				-0.0221 GY	0.41		

46	-0.4342	GY	0.43	0.60	
46				-0.0221 GY	0.63
46				-0.0191 GY	0.68
46				-0.0162 GY	0.74
46				-0.0133 GY	0.79
46				-0.0103 GY	0.85
46				-0.0074 GY	0.90
46				-0.0044 GY	0.95
46				-0.0015 GY	1.00
48				-0.0015 GY	0.04
48				-0.0044 GY	0.08
48				-0.0074 GY	0.14
48				-0.0103 GY	0.19
48				-0.0133 GY	0.25
48				-0.0162 GY	0.30
48				-0.0191 GY	0.35
48				-0.0221 GY	0.41
48				-0.0221 GY	0.46
48				-0.0191 GY	0.51
48				-0.0162 GY	0.57
48				-0.0133 GY	0.62
48				-0.0103 GY	0.68
48				-0.0074 GY	0.73
48				-0.0044 GY	0.78
48				-0.0015 GY	0.83
33				-0.0015 GY	0.04
33				-0.0044 GY	0.08
33				-0.0074 GY	0.14
33				-0.0103 GY	0.19
33				-0.0133 GY	0.25
33				-0.0162 GY	0.30
33				-0.0191 GY	0.35
33				-0.0221 GY	0.41
33	-0.4342	GY	0.43	0.60	
33				-0.0221 GY	0.63
33				-0.0191 GY	0.68
33				-0.0162 GY	0.74
33				-0.0133 GY	0.79
33				-0.0103 GY	0.85
33				-0.0074 GY	0.90
33				-0.0044 GY	0.95
33				-0.0015 GY	1.00
34				-0.0015 GY	0.04
34				-0.0044 GY	0.08
34				-0.0074 GY	0.14
34				-0.0103 GY	0.19
34				-0.0133 GY	0.25
34				-0.0162 GY	0.30
34				-0.0191 GY	0.35
34				-0.0221 GY	0.41
34	-0.4342	GY	0.43	0.60	
34				-0.0221 GY	0.63
34				-0.0191 GY	0.68
34				-0.0162 GY	0.74
34				-0.0133 GY	0.79
34				-0.0103 GY	0.85

STAAD SPACE -- PAGE NO. 20

34			-0.0074 GY	0.90
34			-0.0044 GY	0.95
34			-0.0015 GY	1.00
57			-0.0015 GY	0.04
57			-0.0044 GY	0.08
57			-0.0074 GY	0.14
57			-0.0103 GY	0.19
57			-0.0133 GY	0.25
57			-0.0162 GY	0.30
57			-0.0191 GY	0.35
57			-0.0221 GY	0.41
57			-0.0221 GY	0.46
57			-0.0191 GY	0.51
57			-0.0162 GY	0.57
57			-0.0133 GY	0.62
57			-0.0103 GY	0.68
57			-0.0074 GY	0.73
57			-0.0044 GY	0.78
57			-0.0015 GY	0.83
47			-0.0015 GY	0.04
47			-0.0044 GY	0.08
47			-0.0074 GY	0.14
47			-0.0103 GY	0.19
47			-0.0133 GY	0.25
47			-0.0162 GY	0.30
47			-0.0191 GY	0.35
47			-0.0221 GY	0.41
47	-0.4342 GY	0.43	0.60	
47			-0.0221 GY	0.63
47			-0.0191 GY	0.68
47			-0.0162 GY	0.74
47			-0.0133 GY	0.79
47			-0.0103 GY	0.85
47			-0.0074 GY	0.90
47			-0.0044 GY	0.95
47			-0.0015 GY	1.00
43			-0.0015 GY	0.04
43			-0.0044 GY	0.08
43			-0.0074 GY	0.14
43			-0.0103 GY	0.19
43			-0.0133 GY	0.25
43			-0.0162 GY	0.30
43			-0.0191 GY	0.35
43			-0.0221 GY	0.41
43			-0.0221 GY	0.46
43			-0.0191 GY	0.51
43			-0.0162 GY	0.57
43			-0.0133 GY	0.62
43			-0.0103 GY	0.68
43			-0.0074 GY	0.73
43			-0.0044 GY	0.78
43			-0.0015 GY	0.83
35			-0.0015 GY	0.04
35			-0.0044 GY	0.08
35			-0.0074 GY	0.14
35			-0.0103 GY	0.19

```
 35                          -0.0133 GY   0.25
 35                          -0.0162 GY   0.30
 35                          -0.0191 GY   0.35
 35                          -0.0221 GY   0.41
 35                          -0.0221 GY   0.46
 35                          -0.0191 GY   0.51
 35                          -0.0162 GY   0.57
 35                          -0.0133 GY   0.62
 35                          -0.0103 GY   0.68
 35                          -0.0074 GY   0.73
 35                          -0.0044 GY   0.78
 35                          -0.0015 GY   0.83
 51                          -0.0015 GY   0.04
 51                          -0.0044 GY   0.08
 51                          -0.0074 GY   0.14
 51                          -0.0103 GY   0.19
 51                          -0.0133 GY   0.25
 51                          -0.0162 GY   0.30
 51                          -0.0191 GY   0.35
 51                          -0.0221 GY   0.41
 51    -0.4342 GY   0.43    0.60
 51                          -0.0221 GY   0.63
 51                          -0.0191 GY   0.68
 51                          -0.0162 GY   0.74
 51                          -0.0133 GY   0.79
 51                          -0.0103 GY   0.85
 51                          -0.0074 GY   0.90
 51                          -0.0044 GY   0.95
 51                          -0.0015 GY   1.00
 50                          -0.0015 GY   0.04
 50                          -0.0044 GY   0.08
 50                          -0.0074 GY   0.14
 50                          -0.0103 GY   0.19
 50                          -0.0133 GY   0.25
 50                          -0.0162 GY   0.30
 50                          -0.0191 GY   0.35
 50                          -0.0221 GY   0.41
 50                          -0.0221 GY   0.46
 50                          -0.0191 GY   0.51
 50                          -0.0162 GY   0.57
 50                          -0.0133 GY   0.62
 50                          -0.0103 GY   0.68
 50                          -0.0074 GY   0.73
 50                          -0.0044 GY   0.78
 50                          -0.0015 GY   0.83
 38                          -0.0015 GY   0.04
 38                          -0.0044 GY   0.08
 38                          -0.0074 GY   0.14
 38                          -0.0103 GY   0.19
 38                          -0.0133 GY   0.25
 38                          -0.0162 GY   0.30
 38                          -0.0191 GY   0.35
 38                          -0.0221 GY   0.41
 38    -0.4342 GY   0.43    0.60
 38                          -0.0221 GY   0.63
 38                          -0.0191 GY   0.68
```

STAAD SPACE -- PAGE NO. 22

38			-0.0162 GY	0.74	
38			-0.0133 GY	0.79	
38			-0.0103 GY	0.85	
38			-0.0074 GY	0.90	
38			-0.0044 GY	0.95	
38			-0.0015 GY	1.00	
36			-0.0015 GY	0.04	
36			-0.0044 GY	0.08	
36			-0.0074 GY	0.14	
36			-0.0103 GY	0.19	
36			-0.0133 GY	0.25	
36			-0.0162 GY	0.30	
36			-0.0191 GY	0.35	
36			-0.0221 GY	0.41	
36	-0.4342 GY	0.43	0.60		
36			-0.0221 GY	0.63	
36			-0.0191 GY	0.68	
36			-0.0162 GY	0.74	
36			-0.0133 GY	0.79	
36			-0.0103 GY	0.85	
36			-0.0074 GY	0.90	
36			-0.0044 GY	0.95	
36			-0.0015 GY	1.00	
49			-0.0015 GY	0.04	
49			-0.0044 GY	0.08	
49			-0.0074 GY	0.14	
49			-0.0103 GY	0.19	
49			-0.0133 GY	0.25	
49			-0.0162 GY	0.30	
49			-0.0191 GY	0.35	
49			-0.0221 GY	0.41	
49			-0.0221 GY	0.46	
49			-0.0191 GY	0.51	
49			-0.0162 GY	0.57	
49			-0.0133 GY	0.62	
49			-0.0103 GY	0.68	
49			-0.0074 GY	0.73	
49			-0.0044 GY	0.78	
49			-0.0015 GY	0.83	
53			-0.0015 GY	0.04	
53			-0.0044 GY	0.08	
53			-0.0074 GY	0.14	
53			-0.0103 GY	0.19	
53			-0.0133 GY	0.25	
53			-0.0162 GY	0.30	
53			-0.0191 GY	0.35	
53			-0.0221 GY	0.41	
53	-0.4342 GY	0.43	0.60		
53			-0.0221 GY	0.63	
53			-0.0191 GY	0.68	
53			-0.0162 GY	0.74	
53			-0.0133 GY	0.79	
53			-0.0103 GY	0.85	
53			-0.0074 GY	0.90	
53			-0.0044 GY	0.95	
53			-0.0015 GY	1.00	

```
 48                          -0.0015 GY   0.04
 48                          -0.0044 GY   0.08
 48                          -0.0074 GY   0.14
 48                          -0.0103 GY   0.19
 48                          -0.0133 GY   0.25
 48                          -0.0162 GY   0.30
 48                          -0.0191 GY   0.35
 48                          -0.0221 GY   0.41
 48                          -0.0221 GY   0.46
 48                          -0.0191 GY   0.51
 48                          -0.0162 GY   0.57
 48                          -0.0133 GY   0.62
 48                          -0.0103 GY   0.68
 48                          -0.0074 GY   0.73
 48                          -0.0044 GY   0.78
 48                          -0.0015 GY   0.83
 37                          -0.0015 GY   0.04
 37                          -0.0044 GY   0.08
 37                          -0.0074 GY   0.14
 37                          -0.0103 GY   0.19
 37                          -0.0133 GY   0.25
 37                          -0.0162 GY   0.30
 37                          -0.0191 GY   0.35
 37                          -0.0221 GY   0.41
 37    -0.4342 GY   0.43    0.60
 37                          -0.0221 GY   0.63
 37                          -0.0191 GY   0.68
 37                          -0.0162 GY   0.74
 37                          -0.0133 GY   0.79
 37                          -0.0103 GY   0.85
 37                          -0.0074 GY   0.90
 37                          -0.0044 GY   0.95
 37                          -0.0015 GY   1.00
 50                          -0.0015 GY   0.04
 50                          -0.0044 GY   0.08
 50                          -0.0074 GY   0.14
 50                          -0.0103 GY   0.19
 50                          -0.0133 GY   0.25
 50                          -0.0162 GY   0.30
 50                          -0.0191 GY   0.35
 50                          -0.0221 GY   0.41
 50                          -0.0221 GY   0.46
 50                          -0.0191 GY   0.51
 50                          -0.0162 GY   0.57
 50                          -0.0133 GY   0.62
 50                          -0.0103 GY   0.68
 50                          -0.0074 GY   0.73
 50                          -0.0044 GY   0.78
 50                          -0.0015 GY   0.83
 52                          -0.0015 GY   0.04
 52                          -0.0044 GY   0.08
 52                          -0.0074 GY   0.14
 52                          -0.0103 GY   0.19
 52                          -0.0133 GY   0.25
 52                          -0.0162 GY   0.30
 52                          -0.0191 GY   0.35
```

```
 52                          -0.0221 GY   0.41
 52    -0.4342 GY   0.43    0.60
 52                          -0.0221 GY   0.63
 52                          -0.0191 GY   0.68
 52                          -0.0162 GY   0.74
 52                          -0.0133 GY   0.79
 52                          -0.0103 GY   0.85
 52                          -0.0074 GY   0.90
 52                          -0.0044 GY   0.95
 52                          -0.0015 GY   1.00
 49                          -0.0015 GY   0.04
 49                          -0.0044 GY   0.08
 49                          -0.0074 GY   0.14
 49                          -0.0103 GY   0.19
 49                          -0.0133 GY   0.25
 49                          -0.0162 GY   0.30
 49                          -0.0191 GY   0.35
 49                          -0.0221 GY   0.41
 49                          -0.0221 GY   0.46
 49                          -0.0191 GY   0.51
 49                          -0.0162 GY   0.57
 49                          -0.0133 GY   0.62
 49                          -0.0103 GY   0.68
 49                          -0.0074 GY   0.73
 49                          -0.0044 GY   0.78
 49                          -0.0015 GY   0.83
 39                          -0.0015 GY   0.04
 39                          -0.0044 GY   0.08
 39                          -0.0074 GY   0.14
 39                          -0.0103 GY   0.19
 39                          -0.0133 GY   0.25
 39                          -0.0162 GY   0.30
 39                          -0.0191 GY   0.35
 39                          -0.0221 GY   0.41
 39    -0.4342 GY   0.43    0.60
 39                          -0.0221 GY   0.63
 39                          -0.0191 GY   0.68
 39                          -0.0162 GY   0.74
 39                          -0.0133 GY   0.79
 39                          -0.0103 GY   0.85
 39                          -0.0074 GY   0.90
 39                          -0.0044 GY   0.95
 39                          -0.0015 GY   1.00
 59                          -0.0015 GY   0.04
 59                          -0.0044 GY   0.08
 59                          -0.0074 GY   0.14
 59                          -0.0103 GY   0.19
 59                          -0.0133 GY   0.25
 59                          -0.0162 GY   0.30
 59                          -0.0191 GY   0.35
 59                          -0.0221 GY   0.41
 59                          -0.0221 GY   0.46
 59                          -0.0191 GY   0.51
 59                          -0.0162 GY   0.57
 59                          -0.0133 GY   0.62
 59                          -0.0103 GY   0.68
```

59			-0.0074	GY	0.73
59			-0.0044	GY	0.78
59			-0.0015	GY	0.83
56			-0.0015	GY	0.04
56			-0.0044	GY	0.08
56			-0.0074	GY	0.14
56			-0.0103	GY	0.19
56			-0.0133	GY	0.25
56			-0.0162	GY	0.30
56			-0.0191	GY	0.35
56			-0.0221	GY	0.41
56	-0.4342	GY 0.43	0.60		
56			-0.0221	GY	0.63
56			-0.0191	GY	0.68
56			-0.0162	GY	0.74
56			-0.0133	GY	0.79
56			-0.0103	GY	0.85
56			-0.0074	GY	0.90
56			-0.0044	GY	0.95
56			-0.0015	GY	1.00
57			-0.0015	GY	0.04
57			-0.0044	GY	0.08
57			-0.0074	GY	0.14
57			-0.0103	GY	0.19
57			-0.0133	GY	0.25
57			-0.0162	GY	0.30
57			-0.0191	GY	0.35
57			-0.0221	GY	0.41
57			-0.0221	GY	0.46
57			-0.0191	GY	0.51
57			-0.0162	GY	0.57
57			-0.0133	GY	0.62
57			-0.0103	GY	0.68
57			-0.0074	GY	0.73
57			-0.0044	GY	0.78
57			-0.0015	GY	0.83
40			-0.0015	GY	0.04
40			-0.0044	GY	0.08
40			-0.0074	GY	0.14
40			-0.0103	GY	0.19
40			-0.0133	GY	0.25
40			-0.0162	GY	0.30
40			-0.0191	GY	0.35
40			-0.0221	GY	0.41
40	-0.4342	GY 0.43	0.60		
40			-0.0221	GY	0.63
40			-0.0191	GY	0.68
40			-0.0162	GY	0.74
40			-0.0133	GY	0.79
40			-0.0103	GY	0.85
40			-0.0074	GY	0.90
40			-0.0044	GY	0.95
40			-0.0015	GY	1.00
61			-0.0015	GY	0.04
61			-0.0044	GY	0.08
61			-0.0074	GY	0.14

61			-0.0103 GY	0.19	
61			-0.0133 GY	0.25	
61			-0.0162 GY	0.30	
61			-0.0191 GY	0.35	
61			-0.0221 GY	0.41	
61			-0.0221 GY	0.46	
61			-0.0191 GY	0.51	
61			-0.0162 GY	0.57	
61			-0.0133 GY	0.62	
61			-0.0103 GY	0.68	
61			-0.0074 GY	0.73	
61			-0.0044 GY	0.78	
61			-0.0015 GY	0.83	
55			-0.0015 GY	0.04	
55			-0.0044 GY	0.08	
55			-0.0074 GY	0.14	
55			-0.0103 GY	0.19	
55			-0.0133 GY	0.25	
55			-0.0162 GY	0.30	
55			-0.0191 GY	0.35	
55			-0.0221 GY	0.41	
55	-0.4342 GY	0.43	0.60		
55			-0.0221 GY	0.63	
55			-0.0191 GY	0.68	
55			-0.0162 GY	0.74	
55			-0.0133 GY	0.79	
55			-0.0103 GY	0.85	
55			-0.0074 GY	0.90	
55			-0.0044 GY	0.95	
55			-0.0015 GY	1.00	
59			-0.0015 GY	0.04	
59			-0.0044 GY	0.08	
59			-0.0074 GY	0.14	
59			-0.0103 GY	0.19	
59			-0.0133 GY	0.25	
59			-0.0162 GY	0.30	
59			-0.0191 GY	0.35	
59			-0.0221 GY	0.41	
59			-0.0221 GY	0.46	
59			-0.0191 GY	0.51	
59			-0.0162 GY	0.57	
59			-0.0133 GY	0.62	
59			-0.0103 GY	0.68	
59			-0.0074 GY	0.73	
59			-0.0044 GY	0.78	
59			-0.0015 GY	0.83	
41			-0.0015 GY	0.04	
41			-0.0044 GY	0.08	
41			-0.0074 GY	0.14	
41			-0.0103 GY	0.19	
41			-0.0133 GY	0.25	
41			-0.0162 GY	0.30	
41			-0.0191 GY	0.35	
41			-0.0221 GY	0.41	
41	-0.4342 GY	0.43	0.60		
41			-0.0221 GY	0.63	

STAAD SPACE -- PAGE NO. 27

41			-0.0191 GY	0.68
41			-0.0162 GY	0.74
41			-0.0133 GY	0.79
41			-0.0103 GY	0.85
41			-0.0074 GY	0.90
41			-0.0044 GY	0.95
41			-0.0015 GY	1.00
45			-0.0015 GY	0.04
45			-0.0044 GY	0.08
45			-0.0074 GY	0.14
45			-0.0103 GY	0.19
45			-0.0133 GY	0.25
45			-0.0162 GY	0.30
45			-0.0191 GY	0.35
45			-0.0221 GY	0.41
45			-0.0221 GY	0.46
45			-0.0191 GY	0.51
45			-0.0162 GY	0.57
45			-0.0133 GY	0.62
45			-0.0103 GY	0.68
45			-0.0074 GY	0.73
45			-0.0044 GY	0.78
45			-0.0015 GY	0.83
54			-0.0015 GY	0.04
54			-0.0044 GY	0.08
54			-0.0074 GY	0.14
54			-0.0103 GY	0.19
54			-0.0133 GY	0.25
54			-0.0162 GY	0.30
54			-0.0191 GY	0.35
54			-0.0221 GY	0.41
54	-0.4342 GY	0.43	0.60	
54			-0.0221 GY	0.63
54			-0.0191 GY	0.68
54			-0.0162 GY	0.74
54			-0.0133 GY	0.79
54			-0.0103 GY	0.85
54			-0.0074 GY	0.90
54			-0.0044 GY	0.95
54			-0.0015 GY	1.00
61			-0.0015 GY	0.04
61			-0.0044 GY	0.08
61			-0.0074 GY	0.14
61			-0.0103 GY	0.19
61			-0.0133 GY	0.25
61			-0.0162 GY	0.30
61			-0.0191 GY	0.35
61			-0.0221 GY	0.41
61			-0.0221 GY	0.46
61			-0.0191 GY	0.51
61			-0.0162 GY	0.57
61			-0.0133 GY	0.62
61			-0.0103 GY	0.68
61			-0.0074 GY	0.73
61			-0.0044 GY	0.78
61			-0.0015 GY	0.83

STAAD SPACE -- PAGE NO. 28

```
 42                          -0.0015 GY   0.04
 42                          -0.0044 GY   0.08
 42                          -0.0074 GY   0.14
 42                          -0.0103 GY   0.19
 42                          -0.0133 GY   0.25
 42                          -0.0162 GY   0.30
 42                          -0.0191 GY   0.35
 42                          -0.0221 GY   0.41
 42                          -0.0221 GY   0.46
 42                          -0.0191 GY   0.51
 42                          -0.0162 GY   0.57
 42                          -0.0133 GY   0.62
 42                          -0.0103 GY   0.68
 42                          -0.0074 GY   0.73
 42                          -0.0044 GY   0.78
 42                          -0.0015 GY   0.83
 47                          -0.0015 GY   0.04
 47                          -0.0044 GY   0.08
 47                          -0.0074 GY   0.14
 47                          -0.0103 GY   0.19
 47                          -0.0133 GY   0.25
 47                          -0.0162 GY   0.30
 47                          -0.0191 GY   0.35
 47                          -0.0221 GY   0.41
 47    -0.4342 GY   0.43    0.60
 47                          -0.0221 GY   0.63
 47                          -0.0191 GY   0.68
 47                          -0.0162 GY   0.74
 47                          -0.0133 GY   0.79
 47                          -0.0103 GY   0.85
 47                          -0.0074 GY   0.90
 47                          -0.0044 GY   0.95
 47                          -0.0015 GY   1.00
 58                          -0.0015 GY   0.04
 58                          -0.0044 GY   0.08
 58                          -0.0074 GY   0.14
 58                          -0.0103 GY   0.19
 58                          -0.0133 GY   0.25
 58                          -0.0162 GY   0.30
 58                          -0.0191 GY   0.35
 58                          -0.0221 GY   0.41
 58                          -0.0221 GY   0.46
 58                          -0.0191 GY   0.51
 58                          -0.0162 GY   0.57
 58                          -0.0133 GY   0.62
 58                          -0.0103 GY   0.68
 58                          -0.0074 GY   0.73
 58                          -0.0044 GY   0.78
 58                          -0.0015 GY   0.83
 46                          -0.0015 GY   0.04
 46                          -0.0044 GY   0.08
 46                          -0.0074 GY   0.14
 46                          -0.0103 GY   0.19
 46                          -0.0133 GY   0.25
 46                          -0.0162 GY   0.30
 46                          -0.0191 GY   0.35
```

```
 46                          -0.0221 GY   0.41
 46    -0.4342 GY   0.43    0.60
 46                          -0.0221 GY   0.63
 46                          -0.0191 GY   0.68
 46                          -0.0162 GY   0.74
 46                          -0.0133 GY   0.79
 46                          -0.0103 GY   0.85
 46                          -0.0074 GY   0.90
 46                          -0.0044 GY   0.95
 46                          -0.0015 GY   1.00
 44                          -0.0015 GY   0.04
 44                          -0.0044 GY   0.08
 44                          -0.0074 GY   0.14
 44                          -0.0103 GY   0.19
 44                          -0.0133 GY   0.25
 44                          -0.0162 GY   0.30
 44                          -0.0191 GY   0.35
 44                          -0.0221 GY   0.41
 44                          -0.0221 GY   0.46
 44                          -0.0191 GY   0.51
 44                          -0.0162 GY   0.57
 44                          -0.0133 GY   0.62
 44                          -0.0103 GY   0.68
 44                          -0.0074 GY   0.73
 44                          -0.0044 GY   0.78
 44                          -0.0015 GY   0.83
 54                          -0.0015 GY   0.04
 54                          -0.0044 GY   0.08
 54                          -0.0074 GY   0.14
 54                          -0.0103 GY   0.19
 54                          -0.0133 GY   0.25
 54                          -0.0162 GY   0.30
 54                          -0.0191 GY   0.35
 54                          -0.0221 GY   0.41
 54    -0.4342 GY   0.43    0.60
 54                          -0.0221 GY   0.63
 54                          -0.0191 GY   0.68
 54                          -0.0162 GY   0.74
 54                          -0.0133 GY   0.79
 54                          -0.0103 GY   0.85
 54                          -0.0074 GY   0.90
 54                          -0.0044 GY   0.95
 54                          -0.0015 GY   1.00
 62                          -0.0015 GY   0.04
 62                          -0.0044 GY   0.08
 62                          -0.0074 GY   0.14
 62                          -0.0103 GY   0.19
 62                          -0.0133 GY   0.25
 62                          -0.0162 GY   0.30
 62                          -0.0191 GY   0.35
 62                          -0.0221 GY   0.41
 62                          -0.0221 GY   0.46
 62                          -0.0191 GY   0.51
 62                          -0.0162 GY   0.57
 62                          -0.0133 GY   0.62
 62                          -0.0103 GY   0.68
```

```
 62                          -0.0074 GY   0.73
 62                          -0.0044 GY   0.78
 62                          -0.0015 GY   0.83
 51                          -0.0015 GY   0.04
 51                          -0.0044 GY   0.08
 51                          -0.0074 GY   0.14
 51                          -0.0103 GY   0.19
 51                          -0.0133 GY   0.25
 51                          -0.0162 GY   0.30
 51                          -0.0191 GY   0.35
 51                          -0.0221 GY   0.41
 51    -0.4342 GY   0.43    0.60
 51                          -0.0221 GY   0.63
 51                          -0.0191 GY   0.68
 51                          -0.0162 GY   0.74
 51                          -0.0133 GY   0.79
 51                          -0.0103 GY   0.85
 51                          -0.0074 GY   0.90
 51                          -0.0044 GY   0.95
 51                          -0.0015 GY   1.00
 52                          -0.0015 GY   0.04
 52                          -0.0044 GY   0.08
 52                          -0.0074 GY   0.14
 52                          -0.0103 GY   0.19
 52                          -0.0133 GY   0.25
 52                          -0.0162 GY   0.30
 52                          -0.0191 GY   0.35
 52                          -0.0221 GY   0.41
 52    -0.4342 GY   0.43    0.60
 52                          -0.0221 GY   0.63
 52                          -0.0191 GY   0.68
 52                          -0.0162 GY   0.74
 52                          -0.0133 GY   0.79
 52                          -0.0103 GY   0.85
 52                          -0.0074 GY   0.90
 52                          -0.0044 GY   0.95
 52                          -0.0015 GY   1.00
 62                          -0.0015 GY   0.04
 62                          -0.0044 GY   0.08
 62                          -0.0074 GY   0.14
 62                          -0.0103 GY   0.19
 62                          -0.0133 GY   0.25
 62                          -0.0162 GY   0.30
 62                          -0.0191 GY   0.35
 62                          -0.0221 GY   0.41
 62                          -0.0221 GY   0.46
 62                          -0.0191 GY   0.51
 62                          -0.0162 GY   0.57
 62                          -0.0133 GY   0.62
 62                          -0.0103 GY   0.68
 62                          -0.0074 GY   0.73
 62                          -0.0044 GY   0.78
 62                          -0.0015 GY   0.83
 55                          -0.0015 GY   0.04
 55                          -0.0044 GY   0.08
 55                          -0.0074 GY   0.14
```

55			-0.0103	GY	0.19
55			-0.0133	GY	0.25
55			-0.0162	GY	0.30
55			-0.0191	GY	0.35
55			-0.0221	GY	0.41
55	-0.4342	GY 0.43	0.60		
55			-0.0221	GY	0.63
55			-0.0191	GY	0.68
55			-0.0162	GY	0.74
55			-0.0133	GY	0.79
55			-0.0103	GY	0.85
55			-0.0074	GY	0.90
55			-0.0044	GY	0.95
55			-0.0015	GY	1.00
60			-0.0015	GY	0.04
60			-0.0044	GY	0.08
60			-0.0074	GY	0.14
60			-0.0103	GY	0.19
60			-0.0133	GY	0.25
60			-0.0162	GY	0.30
60			-0.0191	GY	0.35
60			-0.0221	GY	0.41
60			-0.0221	GY	0.46
60			-0.0191	GY	0.51
60			-0.0162	GY	0.57
60			-0.0133	GY	0.62
60			-0.0103	GY	0.68
60			-0.0074	GY	0.73
60			-0.0044	GY	0.78
60			-0.0015	GY	0.83
53			-0.0015	GY	0.04
53			-0.0044	GY	0.08
53			-0.0074	GY	0.14
53			-0.0103	GY	0.19
53			-0.0133	GY	0.25
53			-0.0162	GY	0.30
53			-0.0191	GY	0.35
53			-0.0221	GY	0.41
53	-0.4342	GY 0.43	0.60		
53			-0.0221	GY	0.63
53			-0.0191	GY	0.68
53			-0.0162	GY	0.74
53			-0.0133	GY	0.79
53			-0.0103	GY	0.85
53			-0.0074	GY	0.90
53			-0.0044	GY	0.95
53			-0.0015	GY	1.00
60			-0.0015	GY	0.04
60			-0.0044	GY	0.08
60			-0.0074	GY	0.14
60			-0.0103	GY	0.19
60			-0.0133	GY	0.25
60			-0.0162	GY	0.30
60			-0.0191	GY	0.35
60			-0.0221	GY	0.41
60			-0.0221	GY	0.46

```
     60                          -0.0191 GY   0.51
     60                          -0.0162 GY   0.57
     60                          -0.0133 GY   0.62
     60                          -0.0103 GY   0.68
     60                          -0.0074 GY   0.73
     60                          -0.0044 GY   0.78
     60                          -0.0015 GY   0.83
     56                          -0.0015 GY   0.04
     56                          -0.0044 GY   0.08
     56                          -0.0074 GY   0.14
     56                          -0.0103 GY   0.19
     56                          -0.0133 GY   0.25
     56                          -0.0162 GY   0.30
     56                          -0.0191 GY   0.35
     56                          -0.0221 GY   0.41
     56    -0.4342 GY   0.43    0.60
     56                          -0.0221 GY   0.63
     56                          -0.0191 GY   0.68
     56                          -0.0162 GY   0.74
     56                          -0.0133 GY   0.79
     56                          -0.0103 GY   0.85
     56                          -0.0074 GY   0.90
     56                          -0.0044 GY   0.95
     56                          -0.0015 GY   1.00
     58                          -0.0015 GY   0.04
     58                          -0.0044 GY   0.08
     58                          -0.0074 GY   0.14
     58                          -0.0103 GY   0.19
     58                          -0.0133 GY   0.25
     58                          -0.0162 GY   0.30
     58                          -0.0191 GY   0.35
     58                          -0.0221 GY   0.41
     58                          -0.0221 GY   0.46
     58                          -0.0191 GY   0.51
     58                          -0.0162 GY   0.57
     58                          -0.0133 GY   0.62
     58                          -0.0103 GY   0.68
     58                          -0.0074 GY   0.73
     58                          -0.0044 GY   0.78
     58                          -0.0015 GY   0.83


FOR LOADING -     1
APPLIED JOINT EQUIVALENT LOADS
 JOINT   FORCE-X     FORCE-Y     FORCE-Z      MOM-X       MOM-Y       MOM-Z
    23 0.00000E+00-1.57255E-01 0.00000E+00 1.12696E-02 0.00000E+00-1.75577E-02
    24 0.00000E+00-1.57255E-01 0.00000E+00-1.12696E-02 0.00000E+00-1.75577E-02
    25 0.00000E+00-1.57255E-01 0.00000E+00 1.12696E-02 0.00000E+00 1.75577E-02
    26 0.00000E+00-1.57255E-01 0.00000E+00-1.12696E-02 0.00000E+00 1.75577E-02
    27 0.00000E+00-2.83838E-01 0.00000E+00 1.81003E-02 0.00000E+00-1.68361E-09
    28 0.00000E+00-2.83839E-01 0.00000E+00 1.81003E-02 0.00000E+00-4.10800E-07
    29 0.00000E+00-2.83839E-01 0.00000E+00 1.81003E-02 0.00000E+00 4.04065E-07
    30 0.00000E+00-2.83838E-01 0.00000E+00-1.81003E-02 0.00000E+00-1.17852E-08
    31 0.00000E+00-2.83839E-01 0.00000E+00-1.81003E-02 0.00000E+00-4.02382E-07
    32 0.00000E+00-2.83839E-01 0.00000E+00-1.81003E-02 0.00000E+00 4.14167E-07
    33 0.00000E+00-2.77906E-01 0.00000E+00-3.26619E-07 0.00000E+00-2.87940E-02
```

STAAD SPACE -- PAGE NO. 33

APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
34	0.00000E+00	-2.77906E-01	0.00000E+00	3.27461E-07	0.00000E+00	-2.87940E-02
35	0.00000E+00	-2.77906E-01	0.00000E+00	-3.24094E-07	0.00000E+00	2.87940E-02
36	0.00000E+00	-2.77906E-01	0.00000E+00	3.25778E-07	0.00000E+00	2.87940E-02
37	0.00000E+00	-4.94472E-01	0.00000E+00	-5.50539E-07	0.00000E+00	6.90278E-07
38	0.00000E+00	-4.94472E-01	0.00000E+00	-5.57273E-07	0.00000E+00	-7.00380E-07
39	0.00000E+00	-4.94470E-01	0.00000E+00	-5.55590E-07	0.00000E+00	-1.34688E-08
40	0.00000E+00	-4.94472E-01	0.00000E+00	5.57273E-07	0.00000E+00	6.83544E-07
41	0.00000E+00	-4.94472E-01	0.00000E+00	5.52222E-07	0.00000E+00	-7.03747E-07
42	0.00000E+00	-4.94470E-01	0.00000E+00	5.50539E-07	0.00000E+00	-6.73442E-09

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.207250492E+01
Y = 0.840001988E+02
Z = 0.630251503E+01

TOTAL APPLIED LOAD 1

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -6.41
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 40.40 MY= 0.00 MZ= -13.29

TOTAL REACTION LOAD 1

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 6.41
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -40.40 MY= 0.00 MZ= 13.29

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	0.00000E+00	0
Y =	-5.93946E-01	38
Z =	0.00000E+00	0
RX=	-5.78693E-03	31
RY=	0.00000E+00	0
RZ=	-5.69522E-03	33

STAAD SPACE -- PAGE NO. 34

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
23	0.00	-0.16	0.00	0.01	0.00	-0.02	
	-0.00	-0.24	-0.00	-0.01	-0.00	0.02	111000
24	0.00	-0.16	0.00	-0.01	0.00	-0.02	
	-0.00	0.42	-0.00	0.01	-0.00	0.02	111000
25	0.00	-0.16	0.00	0.01	0.00	0.02	
	-0.00	0.74	-0.00	-0.01	-0.00	-0.02	111000
26	0.00	-0.16	0.00	-0.01	0.00	0.02	
	-0.00	0.37	-0.00	0.01	-0.00	-0.02	111000
29	0.00	-0.28	0.00	0.02	0.00	0.00	
	-0.00	-1.80	-0.00	-0.02	-0.00	-0.00	111000
30	0.00	-0.28	0.00	-0.02	0.00	-0.00	
	-0.00	-0.52	-0.00	0.02	-0.00	0.00	111000
31	0.00	-0.28	0.00	-0.02	0.00	-0.00	
	-0.00	-0.50	-0.00	0.02	-0.00	0.00	111000
32	0.00	-0.28	0.00	-0.02	0.00	0.00	
	-0.00	-0.49	-0.00	0.02	-0.00	-0.00	111000
33	0.00	-0.28	0.00	-0.00	0.00	-0.03	
	-0.00	-0.42	-0.00	0.00	-0.00	0.03	111000
34	0.00	-0.28	0.00	0.00	0.00	-0.03	
	-0.00	-0.39	-0.00	-0.00	-0.00	0.03	111000
35	0.00	-0.28	0.00	-0.00	0.00	0.03	
	-0.00	-0.31	-0.00	0.00	-0.00	-0.03	111000
36	0.00	-0.28	0.00	0.00	0.00	0.03	
	-0.00	-0.40	-0.00	-0.00	-0.00	-0.03	111000

FOR LOADING - 2
APPLIED JOINT EQUIVALENT LOADS

JOINT	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
23	0.00000E+00	-2.24954E-01	0.00000E+00	1.70769E-02	0.00000E+00	-2.80907E-02
24	0.00000E+00	-2.24954E-01	0.00000E+00	-1.70769E-02	0.00000E+00	-2.80907E-02
25	0.00000E+00	-2.24954E-01	0.00000E+00	1.70769E-02	0.00000E+00	2.80907E-02
26	0.00000E+00	-2.24954E-01	0.00000E+00	-1.70769E-02	0.00000E+00	2.80907E-02
27	0.00000E+00	-4.49909E-01	0.00000E+00	3.41538E-02	0.00000E+00	-3.36721E-09
28	0.00000E+00	-4.49911E-01	0.00000E+00	3.41538E-02	0.00000E+00	-7.22267E-07
29	0.00000E+00	-4.49911E-01	0.00000E+00	3.41538E-02	0.00000E+00	7.07114E-07
30	0.00000E+00	-4.49909E-01	0.00000E+00	-3.41538E-02	0.00000E+00	-2.52541E-08
31	0.00000E+00	-4.49911E-01	0.00000E+00	-3.41538E-02	0.00000E+00	-7.00380E-07
32	0.00000E+00	-4.49911E-01	0.00000E+00	-3.41538E-02	0.00000E+00	7.37419E-07
33	0.00000E+00	-4.49906E-01	0.00000E+00	-5.67375E-07	0.00000E+00	-5.61813E-02
34	0.00000E+00	-4.49906E-01	0.00000E+00	5.67375E-07	0.00000E+00	-5.61813E-02
35	0.00000E+00	-4.49906E-01	0.00000E+00	-5.60641E-07	0.00000E+00	5.61813E-02
36	0.00000E+00	-4.49906E-01	0.00000E+00	5.64008E-07	0.00000E+00	5.61813E-02

STAAD SPACE -- PAGE NO. 35

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE DEAD TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.207250507E+01
Y = 0.840001988E+02
Z = 0.630251504E+01

TOTAL APPLIED LOAD 2

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -10.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 68.05 MY= 0.00 MZ= -22.38

TOTAL REACTION LOAD 2

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 10.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -68.05 MY= 0.00 MZ= 22.38

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	0.00000E+00	0
Y =	-1.06176E+00	38
Z =	0.00000E+00	0
RX=	-1.04664E-02	31
RY=	0.00000E+00	0
RZ=	-1.01254E-02	33

EXTERNAL AND INTERNAL JOINT LOAD SUMMARY (KN METE)-

JT	EXT FX/ INT FX	EXT FY/ INT FY	EXT FZ/ INT FZ	EXT MX/ INT MX	EXT MY/ INT MY	EXT MZ/ INT MZ	
							SUPPORT=1
23	0.00	-0.22	0.00	0.02	0.00	-0.03	
	-0.00	-0.40	-0.00	-0.02	-0.00	0.03	111000
24	0.00	-0.22	0.00	-0.02	0.00	-0.03	
	-0.00	0.76	-0.00	0.02	-0.00	0.03	111000
25	0.00	-0.22	0.00	0.02	0.00	0.03	
	-0.00	1.31	-0.00	-0.02	-0.00	-0.03	111000
26	0.00	-0.22	0.00	-0.02	0.00	0.03	
	-0.00	0.66	-0.00	0.02	-0.00	-0.03	111000

STAAD SPACE -- PAGE NO. 36

30	0.00	-0.45	0.00	-0.03	0.00	-0.00	
	-0.00	-0.93	-0.00	0.03	-0.00	0.00	111000
31	0.00	-0.45	0.00	-0.03	0.00	-0.00	
	-0.00	-0.90	-0.00	0.03	-0.00	0.00	111000
32	0.00	-0.45	0.00	-0.03	0.00	0.00	
	-0.00	-0.89	-0.00	0.03	-0.00	-0.00	111000
33	0.00	-0.45	0.00	-0.00	0.00	-0.06	
	-0.00	-0.75	-0.00	0.00	-0.00	0.06	111000
34	0.00	-0.45	0.00	0.00	0.00	-0.06	
	-0.00	-0.71	-0.00	-0.00	-0.00	0.06	111000
35	0.00	-0.45	0.00	-0.00	0.00	0.06	
	-0.00	-0.57	-0.00	0.00	-0.00	-0.06	111000
36	0.00	-0.45	0.00	0.00	0.00	0.06	
	-0.00	-0.72	-0.00	-0.00	-0.00	-0.06	111000

LOAD COMBINATION NO. 101
1.5(DL+LL)

LOADING-	1.	2.
FACTOR -	1.50	1.50

LOAD COMBINATION NO. 201
DL + LL

LOADING-	1.	2.
FACTOR -	1.00	1.00

********** END OF DATA FROM INTERNAL STORAGE **********

64. PARAMETER 1
65. CODE IS800 LSD
66. FYLD 250000 ALL
67. CHECK CODE ALL
STEEL DESIGN

Member Number:	32			
Member Section:	ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)		
Status: PASS	Ratio: 0.121	Critical Load Case: 1	Location:	0.00
Critical Condition: Slenderness (Tension)				

STAAD SPACE -- PAGE NO. 37

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 33					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.402	Critical Load Case: 101	Location:	1.04		
Critical Condition: Sec. 8.2.1.2					

Member Number: 34					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.155	Critical Load Case: 101	Location:	0.00		
Critical Condition: Sec. 8.2.1.2					

Member Number: 35					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.142	Critical Load Case: 101	Location:	0.00		
Critical Condition: Sec. 8.2.1.2					

Member Number: 36					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.436	Critical Load Case: 101	Location:	0.00		
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 38

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 37					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.554	Critical Load Case:	101	Location:	1.04
Critical Condition: Sec. 8.2.1.2					

Member Number: 38					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.592	Critical Load Case:	101	Location:	0.00
Critical Condition: Sec. 8.2.1.2					

Member Number: 39					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.144	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

Member Number: 40					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.144	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

STAAD SPACE -- PAGE NO. 39

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 41					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.144	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

Member Number: 42					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.121	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

Member Number: 43					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.156	Critical Load Case:	101	Location:	0.87
Critical Condition: Sec. 8.2.1.2					

Member Number: 44					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.121	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

STAAD SPACE -- PAGE NO. 40

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 45					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.144	Critical Load Case: 101	Location:	0.87		
Critical Condition: Sec. 8.2.1.2					

Member Number: 46					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.365	Critical Load Case: 101	Location:	1.04		
Critical Condition: Sec. 8.2.1.2					

Member Number: 47					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.216	Critical Load Case: 101	Location:	1.04		
Critical Condition: Sec. 8.2.1.2					

Member Number: 48					
Member Section: ST 50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)				
Status: PASS Ratio: 0.169	Critical Load Case: 101	Location:	0.87		
Critical Condition: Sec. 8.2.1.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 49					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.343	Critical Load Case: 101	Location:	0.87	
Critical Condition: Sec. 8.2.1.2					

Member Number: 50					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.477	Critical Load Case: 101	Location:	0.87	
Critical Condition: Sec. 8.2.1.2					

Member Number: 51					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.144	Critical Load Case: 1	Location:	0.00	
Critical Condition: Slenderness (Tension)					

Member Number: 52					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.219	Critical Load Case: 101	Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 53					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.341	Critical Load Case:	101	Location:	0.43
Critical Condition: Sec. 8.2.1.2					

Member Number: 54					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.144	Critical Load Case:	1	Location:	0.00
Critical Condition: Slenderness (Tension)					

Member Number: 55					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.164	Critical Load Case:	101	Location:	0.43
Critical Condition: Sec. 8.2.1.2					

Member Number: 56					
Member Section: ST	50X50X5.0SHS	(APL APOLLOTUBE SECTIONS)			
Status: PASS	Ratio: 0.214	Critical Load Case:	101	Location:	0.52
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 43

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

Member Number: 57					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS Ratio: 0.282	Critical Load Case: 101		Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

Member Number: 58					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS Ratio: 0.246	Critical Load Case: 101		Location:	0.36	
Critical Condition: Sec. 8.2.1.2					

Member Number: 59					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS Ratio: 0.424	Critical Load Case: 101		Location:	0.00	
Critical Condition: Sec. 8.2.1.2					

Member Number: 60					
Member Section: ST 50X50X5.0SHS		(APL APOLLOTUBE SECTIONS)			
Status: PASS Ratio: 0.386	Critical Load Case: 101		Location:	0.43	
Critical Condition: Sec. 8.2.1.2					

STAAD SPACE -- PAGE NO. 44

STAAD.PRO CODE CHECKING - IS-800-2007-LSD (V2.2)
****

|---|
| Member Number: 61 |
| Member Section: ST 50X50X5.0SHS (APL APOLLOTUBE SECTIONS) |
| Status: PASS Ratio: 0.346 Critical Load Case: 101 Location: 0.00 |
| Critical Condition: Sec. 8.2.1.2 |

|---|
| Member Number: 62 |
| Member Section: ST 50X50X5.0SHS (APL APOLLOTUBE SECTIONS) |
| Status: PASS Ratio: 0.373 Critical Load Case: 101 Location: 0.36 |
| Critical Condition: Sec. 8.2.1.2 |

68. PARAMETER 2
69. CODE IS800 LSD
70. STEEL TAKE OFF ALL

STAAD SPACE -- PAGE NO. 45

STEEL TAKE OFF

STEEL TAKE-OFF

PROFILE	LENGTH(METE)	WEIGHT(KN)
ST 50X50X5.0SHS	29.61	1.901
TOTAL =		1.901

********** END OF DATA FROM INTERNAL STORAGE **********

71. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUL 29,2026 TIME= 18:21:37 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://www.bentley.com/en/support/                      *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*           Copyright (c) Bentley Systems, Inc.             *
*                http://www.bentley.com                     *
*************************************************************
```